M3 Report — G1 GSoC Wi-Fi Sensing

ECE 310 Wireless Communications · Project Milestone 3 (M3) · Team G1

Robust OFDM Channel Modelling for Wi-Fi CSI Sensing in GNU Radio

Experimental validation, root cause at sample level, CSI for sensing, and contribution status for GNU Radio issues #6891 and #7997

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Domain	Wi-Fi Sensing
Mode	GSoC / open source (GNU Radio)
Issues	gnuradio#6891 (packet loss, primary for the contribution), gnuradio#7997 (flowgraph freeze)
Software	GNU Radio 3.10.9.2 (Ubuntu 24.04 package, Python 3.12) for M3; 3.10.12.0 (conda-forge) for M2; E8 independently re-run on a team member's VirtualBox Ubuntu VM (Python 3.14)
Repository	github.com/divyarajv1-ship-it/g1_gsoc_wifisensing → M3_g1_gsoc_wifisensing/

Abstract

GNU Radio's stock OFDM transmitter/receiver silently loses about 4 % of back-to-back packets even on a noise-free channel (#6891), and an OFDM flowgraph can freeze without raising an error (#7997). In M2 we reproduced both, showed the loss is caused by signal processing rather than the scheduler, and proposed an upstream contribution: a multi-packet regression test plus a TX rolloff=4 fix in the official example. In M3 we validate this contribution under conditions M2 did not cover and explain *why* it works. Logging every Schmidl & Cox trigger inside the receiver shows that with the default TX the trigger position jitters over 4–5 samples, while with rolloff=4 it is locked to a single sample. A one-line model of Header/Payload Demux ("a trigger is used only if it arrives at least one frame minus one sample after the previous used trigger") predicts **every lost packet exactly** in all 100 logged runs. Over static multipath (four power-delay profiles), AWGN from 15 dB to noise-free, and carrier frequency offsets up to 3.5 subcarriers, the fix is never worse than the baseline (20/20 conditions) and removes the noise-free loss floor on every profile (e.g. harsh multipath, 30 dB: 88.1 % → 99.9 %; noise-free: 88.5 % → 100 %). For Wi-Fi sensing we extract per-packet CSI from GNU Radio's own channel estimator: its accuracy follows the SNR (–19 dB NMSE at 15 dB, –34 dB at 30 dB), but with the default TX 7–9 % of CSI samples are silently missing, which biases a simple motion-frequency estimate by 7–8 % when timestamps are taken from packet order; with the fix the error is below 0.1 %. Finally, we show that large packets either freeze the flowgraph (300–1000 B, default buffers) or are silently never transmitted (≥ 1020 B, any buffer size), a new limitation relevant to Wi-Fi-MTU-size experiments.

1. Introduction and what changed since M2

M2 delivered: a measurement harness, a quantified reproduction of #6891 (95.9 % PDR on an ideal channel), proof that streaming and offline execution lose identical packets (so the scheduler is not the cause), the counter-intuitive "worse at higher SNR" curve, the buffer/tag mechanism that produces the #7997 freeze symptom, and a tested test+example patch. M2 left open questions, which define M3:

Open question at the end of M2	**Answered in M3 by**
Why exactly does one zero sample fix the loss but one non-zero tail sample (rolloff 2) not fully?	E8 (trigger logging + demux replay model)
Why did 100-byte packets lose nothing?	E8b (packet-length scan; one hypothesis rejected)
Does the fix still hold with multipath and frequency offset, i.e. realistic Wi-Fi channels?	E6, E7

Open question at the end of M2	Answered in M3 by
What does the packet loss mean for Wi-Fi *sensing* (CSI)?	E9
Can Wi-Fi-MTU-size packets (1500 B) be used?	E10
Does the contribution hold on another GNU Radio build?	Contribution re-validation (§6.3)

2. Wireless-communication background used in M3

ECE 310 concept	Where it appears in M3
OFDM, cyclic prefix (CP)	64-subcarrier OFDM, 16-sample CP; every multipath profile is kept shorter than the CP so the channel is a per-subcarrier complex gain
Timing synchronisation	Schmidl & Cox metric; plateau detector trigger position measured to the sample (E8)
Multipath, frequency-selective fading, delay spread	Exponential power-delay profiles with 1, 3, 8 and 14 taps (rms delay ≈ 0, 35, 100, 200 ns at 20 MHz) (E6)
Carrier frequency offset	0–3.5 subcarrier spacings (E7)
SNR, AWGN, BER/packet error	PDR vs SNR curves; uncoded BPSK with CRC32 (E6, E9)
Channel estimation / CSI	ofdm_chanest_vcvc estimate per frame compared with the true H(f) (E9)
Diversity / coding (discussion)	Why uncoded OFDM collapses on frequency-selective channels (§5)

3. Method

3.1 Final method / contribution under evaluation

1. **Upstream contribution (from M2, re-validated here):** test_005_tx_multipacket_rolloff for gr-digital/python/digital/qa_ofdm_txrx.py (100 back-to-back 50-byte packets, random

payloads, rolloff=4, all bytes must match) and rolloff: '4' in gr-digital/examples/ofdm/ofdm_loopback.grc.

2. **Baseline:** stock ofdm_tx (rolloff 0). **Alternatives:** rolloff 1 (GNU Radio resets it to 0), rolloff 2, one zero sample between bursts (burst_shaper_cc).
3. **New M3 tooling** (Code/experiments/m3_harness.py): numpy channel (static or per-burst FIR taps, CFO, AWGN at a defined SNR), receiver instrumentation via ofdm_rx(debug_log=True) to read the Schmidl & Cox trigger stream (sync-detect.dat) and the per-frame channel estimate (channel-estimate.dat), and a replay model of Header/Payload Demux (hpd_replay).

3.2 Evaluation protocol (defined before running)

- Same packets for baseline and fix: 16-bit sequence number + random bytes, fixed seeds 1–5 (1–3 for E8b), 3 unscored flush packets.
- Offline execution (TX run to completion, channel applied, RX run on the stored signal). M2 showed offline and streaming lose the same packets, and offline allows sample-exact bookkeeping.
- Every trial runs in a child process with a timeout; a frozen flowgraph is recorded as HANG.
- **Metrics:** PDR = packets with valid CRC and correct sequence number ÷ packets sent; trigger offset (samples after the true burst start); trigger spacing; frames without CSI; CSI NMSE = $\Sigma(s\cdot|\hat{H}| - |H|)^2 / \Sigma|H|^2$ over the 52 occupied subcarriers with one global real scale factor s; motion-frequency error = $|\hat{f} - f_{true}| / f_{true}$.
- SNR = mean TX sample power (measured from ofdm_tx) ÷ noise power, as in M2.

3.3 Experiments

ID	Question	Varied	Size
E8	At sample level, why are packets lost and why does the fix work?	rolloff 0/1/2/4, gap 1; 20/50/100/200 B	40 runs × 500 packets
E8b	Why do some packet lengths lose nothing?	20 packet lengths, rolloff 0	60 runs × 300 packets
E6	Does the fix hold over multipath + noise?	4 profiles × SNR {15,20,25,30,∞} × rolloff {0,4}	200 runs × 300 packets
E7	Does the fix hold with CFO?	CFO 0–3.5 subcarriers × {flat,	160 runs × 300 packets

ID	Question	Varied	Size
		indoor} × rolloff {0,4}, 25 dB	
E9	What does packet loss do to CSI for sensing?	SNR × rolloff, time-varying indoor channel with a moving reflector (0.5 Hz)	50 runs × 500 packets
E10	Can large (Wi-Fi MTU) packets be used?	100–1500 B × 3 buffer settings	18 runs × 20 packets

4. Results

4.1 E8 — the loss mechanism at sample level

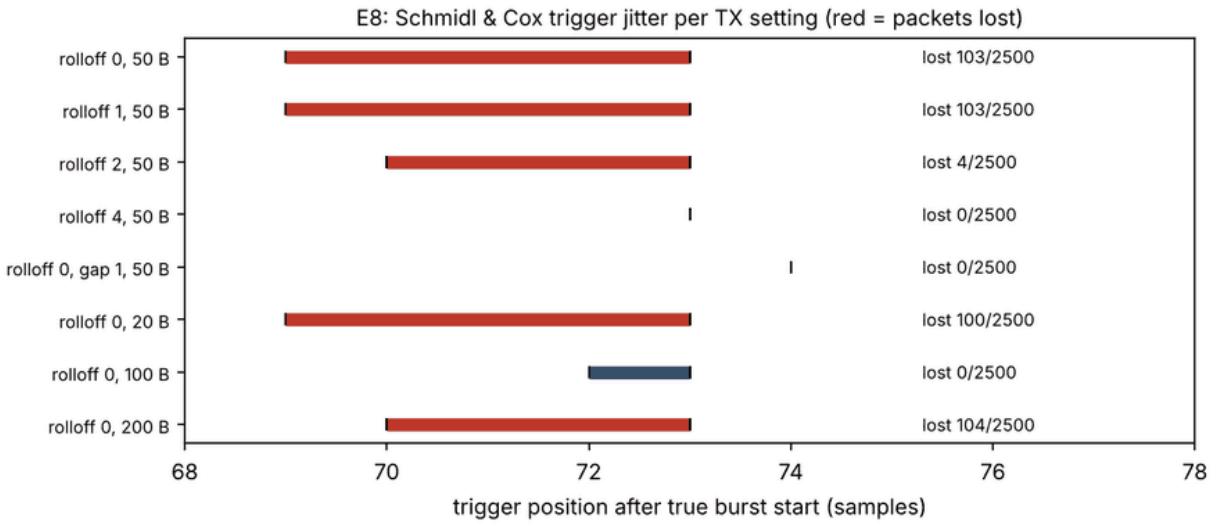


Figure 5. Range of Schmidl & Cox trigger positions relative to the true burst start (5 seeds × 500 packets per row). Red = packets were lost.

TX setting (50 B unless noted)	lost (of 2500)	trigger offset range	trigger spacing	frame / burst samples
rolloff 0 (stock)	103	69–73	956–964	960 / 960

TX setting (50 B unless noted)	lost (of 2500)	trigger offset range	trigger spacing	frame / burst samples
rolloff 1	103	69–73	956–964	960 / 960
rolloff 2	4	70–73	958–964	960 / 961
rolloff 4 (fix)	**0**	**73 only**	**963 only**	960 / 963
gap 1 (zero sample)	0	74 only	961 only	960 / 961
rolloff 0, 20 B	100	69–73	556–564	560 / 560
rolloff 0, 100 B	0	72–73	1679–1681	1680 / 1680
rolloff 0, 200 B	104	70–73	2957–2963	2960 / 2960

Finding 1 — the trigger jitters only when bursts touch. With the stock TX, the plateau detector fires anywhere from 69 to 73 samples after the true start of a burst, because the previous burst's random last symbol leaks into the correlation window. Any separation (rolloff 4's 3 tapered samples, or one zero sample) pins the trigger to one position.

Finding 2 — a one-line model predicts every lost packet. We replayed the logged triggers through the rule *"Header/Payload Demux uses a trigger only if it arrives at least (frame length − 1) samples after the last trigger it used; earlier triggers fall inside the frame it is still consuming and are ignored"*. The packets this rule rejects are **exactly** the lost packets (same sequence numbers) in **40/40** E8 runs and **60/60** E8b runs. This turns the qualitative explanation in the #6891 thread into a quantitative, testable one, and it explains M2's open point: rolloff 2 adds only one sample of slack, so a trigger that is ≥ 3 samples early (rare) still loses the packet (4/2500), while rolloff 4 and gap 1 remove the jitter entirely.

4.2 E8b — why some packet lengths lose nothing

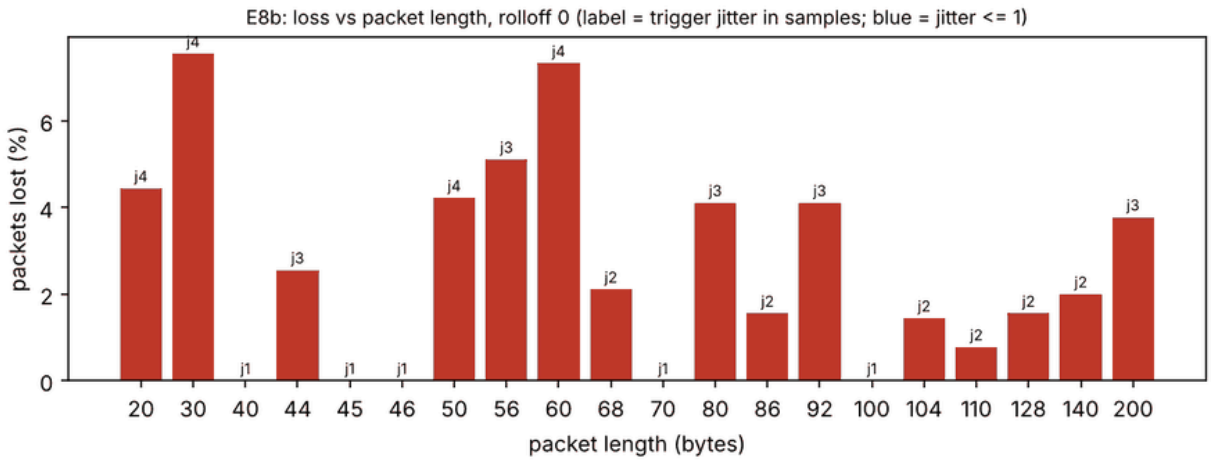


Figure 6. Packets lost vs packet length (rolloff 0, 3 seeds × 300 packets). Label = maximum trigger jitter.

In M3 we tested the hypothesis "loss only happens when the last OFDM payload symbol is completely filled". **It is rejected**: 30 B and 60 B packets have partly filled last symbols and still lose 7–8 % of packets. What *does* hold for all 20 lengths: loss occurs when the trigger jitter is ≥ 2 samples, and no loss when it is ≤ 1 sample (40, 45, 46, 70, 100 B). Why the burst content at these lengths produces a sharper plateau edge is still open; we report it as an observation, not an explanation. The practical consequence: **a single-length test can pass by luck** (e.g. 100 B), which is why the contribution's test uses 50 B and why we add a multi-length check (§6.3).

4.3 E6 — multipath and noise

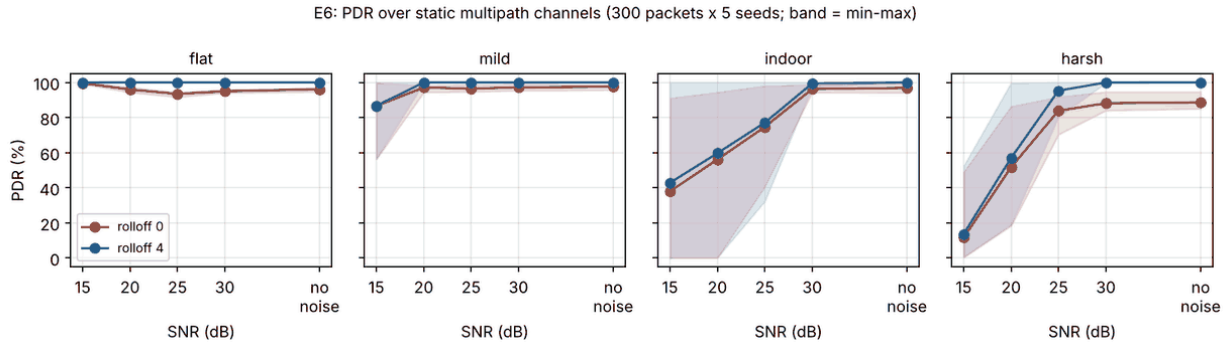


Figure 7. PDR vs SNR for four static multipath profiles; red = stock TX, blue = rolloff 4; band = min–max over 5 seeds.

Profile (rms delay)	SNR 20 dB: stock → fix	SNR 30 dB: stock → fix	Noise-free: stock → fix
flat (0 ns)	95.9 → **100.0**	94.9 → **100.0**	96.1 → **100.0**
mild (≈35 ns)	97.1 → **100.0**	97.1 → **100.0**	97.5 → **100.0**
indoor (≈100 ns)	56.1 → 59.7	96.2 → **99.4**	96.9 → **100.0**
harsh (≈200 ns)	51.7 → 56.9	88.1 → **99.9**	88.5 → **100.0**

(Full table: Results/summary_m3.md.)

- **The fix is never worse.** Rolloff 4 has equal or higher mean PDR in all 20 profile/SNR combinations (gain +0.1 to +11.8 percentage points).
- **Multipath makes the bug worse.** Noise-free, the stock TX loses 3.9 % on a flat channel but **11.5 %** on the harsh profile: a longer channel impulse response smears the previous burst's tail further into the next preamble, increasing the trigger jitter. The fix removes it completely (100 %).
- **At low SNR on frequency-selective channels, both curves collapse** (indoor 15 dB: 38 % vs 43 %). This is not the bug: the stock chain uses *uncoded* BPSK and CRC32, so one deep-faded subcarrier corrupts the packet. Removing that limit needs channel coding / frequency diversity (an ECE 310 concept), which is outside the contribution.

4.4 E7 — carrier frequency offset

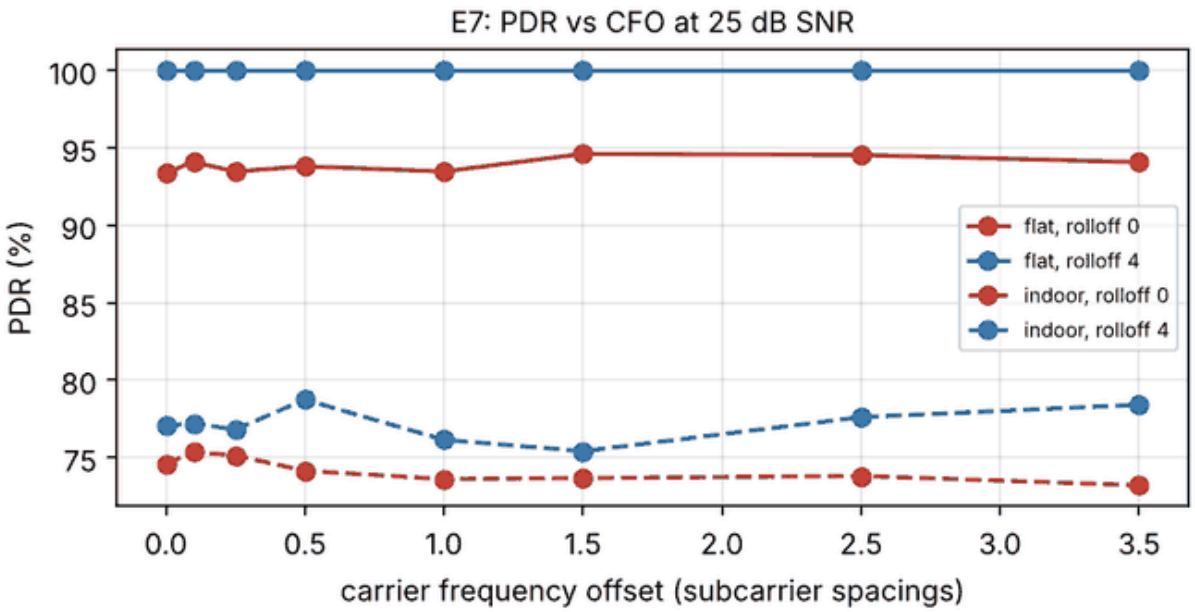


Figure 8. PDR vs CFO at 25 dB SNR.

From 0 to 3.5 subcarrier spacings, PDR does not change for either setting (flat: stock 93.4–94.6 %, fix 100 % at every CFO; indoor: stock 73.2–75.3 %, fix 75.4–78.7 %). GNU Radio's Schmidl & Cox fine/coarse CFO estimation handles these offsets, and the timing problem is independent of CFO. The fix's benefit is therefore robust to oscillator mismatch.

4.5 E9 — what the loss means for Wi-Fi sensing (CSI)

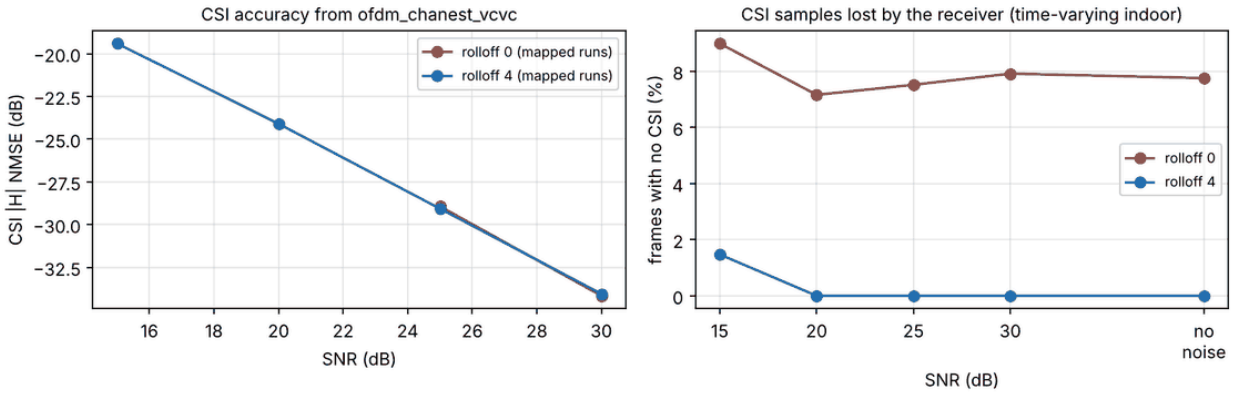


Figure 9. Left: accuracy of the per-frame CSI from ofdm_chanest_vcvc vs SNR. Right: fraction of frames for which the receiver produced no CSI at all.

Set-up: time-varying indoor channel (8 static taps + one reflector path whose gain and phase oscillate at 0.5 Hz, like a person moving), 50 packets/s, 500 packets.

SNR	**setting**	**frames with no CSI (of 503)**	**CSI NMSE**	**motion-freq. error, timestamps from sequence numbers**	**motion-freq. error, timestamps from packet order**
25 dB	stock	37.8 (7.5 %)	−28.9 dB	0.00 %	**8.31 %**
25 dB	fix	0	−29.1 dB	0.06 %	0.06 %
30 dB	stock	39.8 (7.9 %)	−34.2 dB	0.00 %	**7.63 %**
30 dB	fix	0	−34.1 dB	0.06 %	0.06 %
noise-free	stock	39.0 (7.8 %)	< −60 dB	0.00 %	**7.40 %**

SNR	setting	frames with no CSI (of 503)	CSI NMSE	motion-freq. error, timestamps from sequence numbers	motion-freq. error, timestamps from packet order
noise-free	fix	0	< −60 dB	0.00 %	0.00 %

- **GNU Radio already computes good CSI.** The estimate from ofdm_chanest_vcvc tracks the true |H(f)| with an NMSE that falls about 1 dB per dB of SNR (−19 dB at 15 dB → −34 dB at 30 dB), and is exact without noise. But ofdm_rx does **not** expose it: the CSI tag ofdm_sync_chan_taps does not reach the receiver's output (we checked the output tags: only packet_num, ofdm_sync_carr_offset and the length tag arrive). We had to use the debug file sinks.
- **The bug removes 7–9 % of CSI samples** on this time-varying multipath channel, about twice the 4 % payload loss on a flat channel, consistent with E6. With the fix, no CSI frame is missing at ≥ 20 dB.
- **Silent gaps bias sensing.** If the sensing pipeline assumes CSI samples are evenly spaced (as many CSI tools do when drops are not reported), the time axis is compressed and the estimated motion frequency is 7–8 % too high. With timestamps recovered from sequence numbers, or with the fix, the error is < 0.1 %. This is the concrete link between #6891 and Wi-Fi sensing that our project set out to show.
- Caveat: with the stock TX at ≤ 25 dB, CSI frames could not be matched one-to-one to packets in 14 of 15 runs (noise makes some triggers marginal), so the stock-TX NMSE and frequency values come from 1 run (25 dB), 3 runs (30 dB) and 5 runs (noise-free); cells with no matched run are not reported.

4.6 E10 — large packets and the freeze (#7997 mechanism)

Packet size	default buffers	ofdm_tx/ofdm_rx buffers = 65536	+ source block 65536
100, 250 B	OK 20/20	OK 20/20	OK 20/20
300 B	**HANG** (ofdm_cyclic_prefixer : Buffer too small for min_noutput_items)	OK 20/20	OK 20/20
600, 1000 B	**HANG** (tagged_stream_mux : Buffer too small …)	OK 20/20	OK 20/20

Packet size	**default buffers**	**ofdm_tx/ofdm_rx buffers = 65536**	**+ source block 65536**
1500 B	finishes, **0/20 received**	0/20	0/20

- The freeze mechanism from M2 reproduces on a second GNU Radio build (3.10.9.2): the failing block's thread exits, the rest of the flowgraph waits forever, run() never returns.
- Enlarging the hier blocks' output buffers fixes sizes up to **1019 bytes** (verified 1000–1019 B: 20/20).
- **New finding:** for ≥ **1020 bytes** (≥ 8192 payload bits with CRC) the TX's tagged_stream_mux logs "requesting more input data than we can provide. ninput_items_required = 12032, max_possible_items_available = 8191" (1500 B), outputs nothing, and the flowgraph *ends normally* with zero packets and no exception, even with 65536-item buffers. Wi-Fi-MTU packets therefore cannot be sent through the stock OFDM chain without changes; this is a third silent-failure mode of the same family (a block silently refusing work) and a candidate follow-up issue.

5. Critical analysis

What worked. The contribution does what it claims, under more conditions than M2 tested: ideal, AWGN, four multipath profiles, CFO up to 3.5 subcarriers, two GNU Radio builds, four packet lengths. The mechanism is now explained to the sample and predicted exactly, which is the strongest evidence we can give maintainers that rolloff=4 is not a lucky constant: any separation that pins the trigger works, and rolloff 4 is the smallest standard setting that does so.

What did not work / limits.

1. *The fix is a workaround at the TX.* A receiver that receives back-to-back bursts from another transmitter (e.g. real hardware) is still vulnerable. A receiver-side fix (letting Header/Payload Demux accept a trigger that falls in the last few samples of the current frame, or refining timing after detection) would be more general but changes C++ behaviour and needs maintainer discussion.
2. *Low-SNR frequency-selective performance* is dominated by uncoded BPSK; our fix cannot help there (E6, indoor/harsh ≤ 20 dB).
3. *E8b is only partly explained*: we know jitter predicts loss, but not why certain lengths produce no jitter. One hypothesis was rejected.
4. *Simulation only.* Channels are synthetic (exponential PDPs, a sinusoidal reflector), not measured. Real SDR hardware adds sample-rate offset, phase noise and AGC effects.
5. *#7997 itself is not confirmed*: the reporter's flowgraph is not public; we show a mechanism producing the same symptom, now on two GNU Radio versions.

6. *CSI mapping at low SNR with the stock TX* was not always possible (E9 caveat).
7. *Large packets*: buffers fix up to 1019 B only; ≥ 1020 B fails silently.

Why the results make sense. The loss floor rises with SNR (M2) and with delay spread (E6) for the same reason: both make the previous burst's tail more coherent in the next preamble's correlation window, so the trigger moves early systematically rather than randomly.

6. Final contribution and its status

6.1 What it changes

File (gnuradio/gnuradio)	**Change**
gr-digital/python/digital/qa_ofdm_txrx.py	+ _txrx_multipacket() helper, + test_005_tx_multipacket_rolloff (100 × 50 B, rolloff 4) — first multi-packet OFDM test upstream
gr-digital/examples/ofdm/ofdm_loopback.grc	rolloff: '0' → rolloff: '4'

Contribution level (course §5.2): **Level C (testing)** with a **Level A/B** example fix; E8/E10 provide the evidence base for a Level B/D follow-up.

6.2 Technical significance

- Closes a real coverage gap (all existing qa_ofdm_txrx tests send one packet; #6891 only appears between packets).
- The test is shown to fail on the bug and pass with the fix (negative control).
- The example is the entry point for most GNU Radio OFDM users and for students; it currently loses ~4 % of packets silently and up to 11.5 % on multipath (E6).
- No C++ or API change, small diff, easy to review; backed by 500+ measured runs across M2+M3.

6.3 Testing / validation evidence

$ python Code/contribution/test_multipacket_regression.py # GNU Radio 3.10.9.2

test_005_tx_multipacket_rolloff ... ok

test_negative_control_rolloff0_detects_bug ... ok (stock TX loses data -> test would fail)

test_rolloff4_many_lengths_and_seeds ... ok (20/50/100/200 B x 3 seeds)

Ran 3 tests in 1.289s OK

M2: same test on 3.10.12.0 — 6/6 qa_ofdm_txrx tests OK; with rolloff 0 the new test fails (4850 != 5000).
Independent re-run on a second machine (01/11/2026, team member's VirtualBox Ubuntu VM, Python 3.14): E8 was re-run in full with the same scripts and seeds (40 runs, Results/e8_trigger_timing.csv). Every lost-packet count, trigger-offset range and trigger spacing is identical to §4.1 (e.g. stock TX, 50 B, seeds 1–5: 26, 21, 19, 18, 19 lost packets), and the replay model predicted the lost packets exactly in 40/40 runs. The first four E8b runs (20 B seeds 1–3 and 30 B seed 1: 18, 12, 10 and 26 lost) were also identical before that run was stopped. E6, E7, E9 and E10 were run once, on GNU Radio 3.10.9.2.

6.4 Status of the upstream process (course §5.3 workflow)

Step	**Status**
Explore repository, CONTRIBUTING.md, GREP1	Done (M2)
Select issue	Done
Discuss with maintainers (issue comments on #6891 / #7997)	Drafted in M2 (M2_g1_gsoc_wifisensing/Code/upstream_patch/ISSUE_COMMENTS.md); not yet posted as of 01/11/2026
Branch + implementation + tests	Done; patch in M2_g1_gsoc_wifisensing/Code/upstream_patch/, re-validated in M3_g1_gsoc_wifisensing/Code/contribution/test_multipacket_regression.py (not yet pushed to a gnuradio fork)
Pull request	Not yet opened; planned for M4 after posting the issue comment and testing on a source build of main
Review / merge	`Not applicable yet (no PR open)`

7. Maintainer and community feedback

Date	Who	What	Our response
Feb 2024	marcusmueller (maintainer)	Confirmed #6891: "a rather serious" bug	We quantify it and provide a test
Jun 2026	ffu (contributor)	Plateau + demux explanation; recommends TX rolloff as least-invasive change	E8 confirms it to the sample and gives a predictive model; our PR implements the recommendation
Jun 2026	Wiktorijus	Suspected buffers/scheduler	M2 E1 (streaming = offline) and M3 E8 rule this out
9 Oct 2026	vamsip-cell	Offered to investigate #7997	We coordinate rather than duplicate
01/11/2026	—	No reply to us yet: our comments on #6891 / #7997 have not been posted	Post our comments and follow up on the GNU Radio Matrix chat (planned for M4)

8. Revised SOTA positioning (M2 → M3)

Project / work	What it does	Our position after M3 evidence
GNU Radio 3.10 gr-digital OFDM	Generic OFDM TX/RX used in examples and teaching	We now know three silent-failure modes (timing-jitter drop, buffer freeze, ≥ 1020 B refusal), have a predictive model of the first, and a validated fix + test for it. Upstream has none of this documented or tested.
GNU Radio 4.0	New runtime / buffer model	The drop (#6891) is signal processing, so it will **also**

Project / work	What it does	Our position after M3 evidence
		exist in GR4 unless the blocks change; the freeze/refusal modes depend on 3.x buffers and may not. Our tests are portable checks for GR4. *(Changed from M2, where we only expected the freeze class to be runtime-specific.)*
gr-ieee802-11 (Bloessl)	Standard 802.11a/g/p PHY	Uses its own sync; our trigger-logging + replay method is directly applicable to measure its packet-level reliability.
CSI tools (Intel 5300, Atheros, Nexmon, ESP32-CSI, PicoScenes)	CSI from commercial chips	Real hardware but closed PHY. M3 shows GNU Radio already computes accurate CSI (−34 dB NMSE at 30 dB) but does not expose it, and that silent drops bias sensing. A controlled, open CSI source is feasible — the basis of our GSoC 2027 proposal.
SenseFi (Yang et al., 2023)	Datasets + deep models for CSI sensing	Assumes uniformly sampled CSI; E9 quantifies how a 7–9 % silent drop violates that assumption (7–8 % frequency bias).
IEEE 802.11bf (Du et al.)	WLAN sensing procedures	Requires reliable, timestamped sensing measurements; our results argue for exposing CSI *with* packet numbers/timestamps.

Project / work	What it does	Our position after M3 evidence
GSoC 2026 GNU Radio projects (HIL CI, gr-cyberether, GR4 remote plotting)	Testing infra, visualisation	Our multi-packet PDR test is a ready HIL experiment; CSI output could feed their plotting tools.

Where G1 stands now. We do not claim a new algorithm. Our contribution is better-tested and now *explained* infrastructure one layer below the sensing literature, with the first quantitative link between OFDM receiver reliability in GNU Radio and CSI-sensing error. The remaining distance to SOTA is: (1) getting the fix merged upstream, (2) a receiver-side fix, (3) a supported CSI output in ofdm_rx, and (4) validation on SDR hardware.

9. Individual contributions

Member	M3 responsibility	Evidence (commit / file)
Hirva Vekariya	Docs, GSoC 2027 proposal draft, report source (Docs/, Proposal/, Report/*.md)	Commit link: https://github.com/divyarajv1-ship-it/g1_gsoc_wifisensing/commit/ef0cb7d Member's own comment: M3: docs and GSoC 2027 proposal draft
Tirth Pathar	Multipath/SNR and CFO sweeps E6, E7 (e6_multipath_snr.py, e7_cfo.py)	Commit link: https://github.com/divyarajv1-ship-it/g1_gsoc_wifisensing/commit/eec3343 Member's own comment: M3: multipath/SNR and CFO sweeps E6/E7
Divyaraj Vaghela	Repository owner; harness and timing experiments E8, E8b (m3_harness.py, e8_*.py), README	Commit link: https://github.com/divyarajv1-ship-it/g1_gsoc_wifisensing/commit/56e0fe3de409190a166d3fa34c7b0b4fc7d6efaa

Member	M3 responsibility	Evidence (commit / file)
		Member's own comment: M3: folder, README, harness and timing experiments E8/E8b
Jahnavi Patel	Contribution re-test, analysis script and results (Code/contribution/, Code/analysis/, Results/)	Commit link: https://github.com/divyarajv1-ship-it/g1_gsoc_wifisensing/commit/2f0ff7342108b5f5b4e0bfce597e2b8fdc07cf18 Member's own comment: M3: contribution re-test, analysis script and results
Jinay Shah	CSI-for-sensing experiment E9 and large-packet test E10 (e9_csi_sensing.py, e10_large_packets.py)	Commit link: https://github.com/divyarajv1-ship-it/g1_gsoc_wifisensing/commit/911c090 Member's own comment: M3: CSI-for-sensing experiment E9 and large-packet test E10

10. Limitations and future work (towards M4)

1. Post the issue comments and open the PR (if not done); respond to review.
2. Propose (issue first) a receiver-side tolerance in Header/Payload Demux and a "fail loudly" check for oversized frames.
3. Report the ≥ 1020 B silent refusal as a new issue with e10_large_packets.py as reproducer.
4. Expose per-packet CSI from ofdm_rx (optional output or message port) — GSoC 2027 core idea.
5. Repeat E6/E9 on SDR hardware (e.g. two PlutoSDRs / USRPs) if available.

11. Reproducibility

M3_g1_gsoc_wifisensing/Code/run_all.sh re-runs everything (≈ 25 min); Code/environment.yml (or apt install gnuradio on Ubuntu 24.04); raw CSVs, logs and figures in Results/; fixed seeds everywhere. Results/summary_m3.md is regenerated from the CSVs by

Code/analysis/make_m3_figures.py. On 01/11/2026 a team member re-ran E8 with run_all.sh on a VirtualBox Ubuntu VM (Python 3.14) and obtained identical results (§6.3).

12. Use of generative AI

We used an AI assistant (Claude, by Anthropic) to help write the M3 experiment scripts and analysis code, to run the first full set of experiments (GNU Radio 3.10.9.2), and to draft this report, the README and the GSoC 2027 proposal draft. A team member independently re-ran experiment E8 on their own VirtualBox Ubuntu VM and obtained identical numbers (§6.3). Every team member has read the code and this report, and each of us can explain every equation, script, figure and design decision. No result in this report was produced without running the corresponding script, and no maintainer feedback or upstream status has been invented.

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